GALILEAN MOON

Io

DISTANCE FROM JUPITER 261,800 miles (421,600 km)

ORBITAL PERIOD 1.77 Earth days

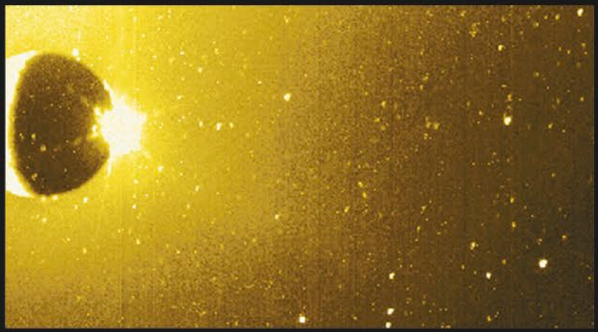
DIAMETER 2,262 miles (3,643 km)

Io is a little larger and denser than Earth's Moon and orbits Jupiter at a distance only slightly greater than the Moon's from Earth. But there the similarities end. Io is a highly colored world of volcanic pits, calderas and vents, lava flows, and high-reaching plumes. The moon's nature was revealed first by the two Voyager probes and then more fully explored by the Galileo mission. Prior to Voyager 1's arrival in March 1979, scientists expected to find a cold, impact-cratered moon. Instead, it found the most volcanic body in the solar system.

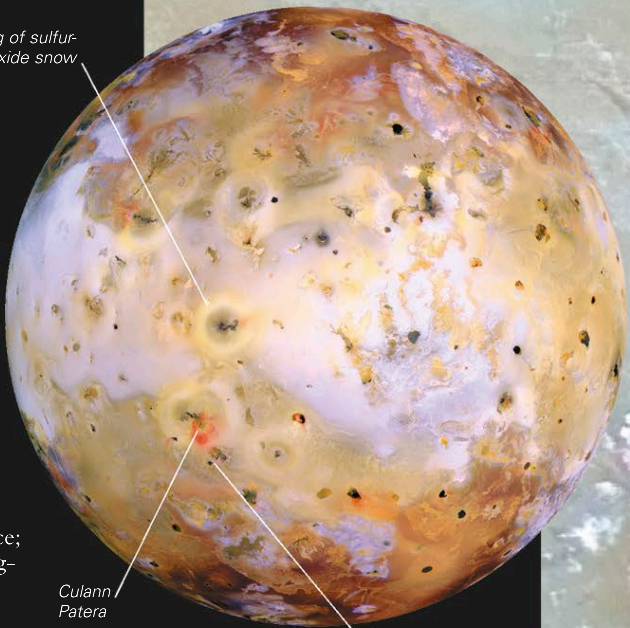
Io has a thin silicate crust that surrounds a molten silicate layer. Below this lies a comparatively large iron-rich core that extends about halfway to the surface. Io orbits Jupiter quickly, every 42.5 hours or so. As it orbits, it is subjected to the strong gravitational pull of Jupiter on one side and the lesser pull of Europa on the other. Io's surface flexes as a consequence of the varying strength and direction of the pull it experiences. The flexing is accompanied by friction, which produces the heat that keeps part of Io's interior molten. It is this material that erupts through the surface and constantly renews it.

The evidence of such volcanism is seen all over Io. Over 80 major active volcanic sites and more than 300 vents have been identified. Features known as plumes are also found at the surface; these fast-moving and long-lived columns of cold gas and frost grains are more like geysers than volcanic explosions. They are created as superheated sulfur dioxide shoots through fractures in Io's crust. The material in the plumes falls slowly back to the surface as snow and leaves circular or oval frost deposits. Plume material also spreads into space surrounding Io and supplies a doughnut-shaped body of material that has formed along Io's orbital path. Temperatures at the volcanic hot spots can be over 2,240°F (1,230°C), the highest surface temperatures in the solar system outside the Sun. Elsewhere, the surface is cold, reaching just -244°F (-153°C).

Simon Marius (see p.182) suggested the names of the Galilean moons. Io is named after one of Zeus's loves, whom he changed into a cow to hide her from his jealous wife. Hera was not fooled and sent a gadfly to torment Io forever. Other surface features are named after people and places from the Io myth or from Dante's *Inferno*, or after fire, sun, volcano, and thunder gods, goddesses, and heroes.



JUPITERSHINE
Sunlight reflected off Jupiter illuminates Io's western side. The eastern side is in shadow except for a burst of light beyond the limb where the plume of the volcano Prometheus is lit. The yellowish sky is produced by sodium atoms surrounding Io scattering the sunlight.



VOLCANIC ACTIVITY
In this color-enhanced Galileo image, the dark spots on Io's surface are active volcanic centers. The dark, eruptive area of Prometheus at center left is encircled by a pale yellow ring of sulfur-dioxide snow deposited by the volcano's plume.

TOHIL MONS
Nonvolcanic mountains are also found on Io. Here, the sunlit peak of the 185-mile (300-km) wide Tohil Mons rises 3.4 miles (5.4 km) above Io's surface.

CULANN PATERA
Colorful lava flows stream away from the irregularly shaped green-floored volcanic crater of Culann Patera (right of center). The reasons for the varied colors are uncertain. The diffuse red material is thought to be a compound of sulfur deposited from a plume of gas. The green deposits may be formed when sulfur-rich material coats warm silicate lava.

THE SOLAR SYSTEM